

Distributed Control of Multi-Robot Systems with Global Connectivity Maintenance

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Abstract—In this paper we present a decentralized control strategy for the connectivity maintenance for groups of mobile robots performing some desired task. Exploiting a completely decentralized estimation strategy for the algebraic connectivity of the graph, we prove the connectivity maintenance property in the general case, i.e. in presence of a generic (bounded) additional control term. Then, we address two specific decentralized control applications: rendezvous and formation control. We analytically prove that our control strategy ensures the global connectivity maintenance, while preserving the convergence properties of the rendezvous and formation controllers respectively. Simulations and experimental results are presented as well, in order to show the effectiveness of the proposed control law.

I. INTRODUCTION

In this paper we address the connectivity maintenance issue for distributed control applications. In [13] we introduced a connectivity maintenance control strategy based on the decentralized estimation of the algebraic connectivity of the communication graph. This control strategy extends the results first introduced in [17], providing a method to probably guarantee connectivity maintenance in multi-agent systems. In this paper we extend the results provided in [13], taking explicitly into account the presence of external control laws. More specifically, we are interested in guaranteeing that the robots keep the connectivity *while* performing some desired cooperative task. We show that, provided that the desired control law is bounded, our strategy ensures the maintenance of the connectivity. In particular, we refer to two specific examples of cooperative tasks: rendezvous and formation control.

Decentralized control of groups of mobile robots has several applications, such as surveillance, exploration of unknown environments, search-and-rescue, automatic warehouses. In this kind of problems, robots are supposed to coordinate their motion, in order to achieve the global objective. Generally speaking, one of the main challenges is due to the fact that robots move into cluttered environments: unknown terrains to be explored, industrial environments for goods transportation, damaged buildings in search-and-rescue applications. This implies that collisions with obstacles must be avoided, while robots are performing their

task. Thus, obstacle avoidance can interfere with the primary task of the robots: problems can arise, for instance, if some robots are trapped. One of the main issues, in this case, is to ensure the connectivity maintenance: due to the limited communication capabilities, in fact, the trapped robots can lose the connectivity with the rest of the group, which implies that the global objective can not be, in general, achieved. In this paper, we analyze two different global objectives: rendezvous and formation control. In the simulations described in Section VI we provide some examples where, due to the presence of obstacles, robots lose connectivity, while performing rendezvous or formation control. We then provide a control strategy to ensure the connectivity maintenance.

In the literature, several approaches to connectivity maintenance have been proposed. These approaches can be divided into two categories: approaches to maintain the local connectivity, and approaches to maintain the global connectivity.

Maintaining the local connectivity entails designing a controller that ensures that, if a communication link is active at time $t = 0$, it will be active $\forall t \geq 0$. Examples of decentralized algorithms for local connectivity maintenance can be found for instance in [2], [3], [9], and references therein.

Nevertheless, to ensure that information exchange among all the robots is possible, it is necessary to guarantee only the *global* connectivity of the communication graph. Loosely speaking, it is acceptable that a few links are broken, as long as the overall graph is still connected: if necessary, redundant links can be removed, and new ones can be introduced. Thus, imposing the global connectivity maintenance ensures that none of the robots loses connectivity from the rest of the group, while not reducing the ability of the group to move into cluttered environments.

The control strategy introduced in [13] ensures the maintenance of the global connectivity. More specifically, inspired by [17], a decentralized estimation procedure has been defined to compute the value of the second-smallest eigenvalue of the Laplacian matrix that, as shown in [6], is a measure of the connectivity of a graph. Then, a decentralized control strategy has been designed that guarantees maintenance of the connectivity. With respect to other approaches that can be found in the literature (e.g. [4], [18], [10]), one of the main advantages of the estimation procedure defined in [17] is that it provides an estimate of the gradients of the Fiedler eigenvalue, as well as the eigenvalue itself.

The outline of the paper is as follows. In Section II we provide some background on graph theory. In Section III we introduce the connectivity maintenance control strategy.

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In Section IV we provide the main result of the paper, i.e. the prove of connectivity maintenance in the presence of a bounded external control term. In Section V we provide a solution for the connectivity maintenance issue in the rendezvous and formation control problems, respectively. Simulations and experiments are described in Section VI, and concluding remarks are provided in Section VII.

II. BACKGROUND ON GRAPH THEORY

In this section we summarize some of the main notions on graph theory used in the paper. Further details can be found for instance in [8]. Given N mobile robots, we describe the communication architecture among them as an undirected graph. Each robot corresponds to a node of the graph, and each link between two robots corresponds to an edge of the graph. Let $\mathcal{N}_i$ be the neighborhood of the i -th robot, i.e. the set of robots that can exchange information with the i -th one. The communication graph can be described by means of the adjacency matrix $A \in \mathbb{R}^{N \times N}$. Each element a_{ij} is defined as the weight of the edge between the i -th and the j -th robot, and is a positive number if $j \in \mathcal{N}_i$, zero otherwise. Since we are considering undirected graphs, we assume $a_{ij} = a_{ji}$. The degree matrix of the graph is defined as $D = \text{diag}(\{d_i\})$, where d_i is the degree of the i -th node of the graph, i.e. $d_i = \sum_{j=1}^N a_{ij}$. The (weighted) Laplacian matrix of the graph is defined as $L = D - A$. The unweighted Laplacian matrix, L_* , is defined as a special case of Laplacian matrix, where all non-zero entries of the adjacency matrix are equal to one. The Laplacian matrix exhibits some remarkable properties:

- 1) Let $\mathbf{1}$ be the column vector of all ones. Then, $L\mathbf{1} = \mathbf{0}$.
- 2) Let $\lambda_i, i = 1 \dots, N$ be the eigenvalues of the Laplacian matrix.

- The eigenvalues can be ordered such that

$$0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N \quad (1)$$

- $\lambda_2 > 0$ if and only if the graph is connected: then, λ_2 is defined as the algebraic connectivity of the graph.
- Thus, if the graph is connected, $\ker(L) = \text{span}(\mathbf{1})$.

III. CONNECTIVITY MAINTENANCE AND ESTIMATION PROCEDURE

A. Connectivity maintenance control strategy

In this paper we consider a group of N single integrator agents, i.e.:

$$\dot{p}_i = u_i \quad (2)$$

where $p_i \in \mathbb{R}^m$ is the position of the i -th agent, and u_i is the control input. Let $p = [p_1^T \dots p_N^T]^T \in \mathbb{R}^{Nm}$ be the state vector of the multi-agent system.

In [13] we introduced the following control law:

$$\dot{p}_i = u_i^c \quad (3)$$

where u_i^c is defined as follows:

$$u_i^c = \text{csch}^2(\lambda_2 - \epsilon) \frac{\partial \lambda_2}{\partial p_i} \quad (4)$$

where ϵ is the desired lower-bound for the value of λ_2 . As shown in Fig. 1(a), the magnitude of the control action suddenly increases as the connectivity of the graph worsens.

Let R be the maximum communication range for each agent, i.e. the j -th agent is inside $\mathcal{N}_i$ if $\|p_i - p_j\| \leq R$. The edge-weights a_{ij} are defined as follows:

$$a_{ij} = \begin{cases} e^{-(\|p_i - p_j\|^2)/(2\sigma^2)} & \text{if } \|p_i - p_j\| \leq R \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

The scalar parameter σ is chosen to satisfy the threshold condition $e^{-(R^2)/(2\sigma^2)} = \Delta$, where Δ is a small predefined threshold.

Let v_2 be the eigenvector corresponding to the eigenvalue λ_2 . Given the definition of the edge-weights in Eq. (5), the value of $\frac{\partial \lambda_2}{\partial p_i}$ can be computed as shown in [17]:

$$\frac{\partial \lambda_2}{\partial p_i} = \sum_{j \in \mathcal{N}_i} -a_{ij} (v_2^i - v_2^j)^2 \frac{p_i - p_j}{\sigma^2} \quad (6)$$

where v_2^k is the k -th component of v_2 .

B. Estimation of the algebraic connectivity of the graph

The computation of the eigenvectors of the Laplacian matrix is a centralized operation. Hence, the actual values of λ_2 , v_2 and $\frac{\partial \lambda_2}{\partial p_i}$ are not available to the agents. As shown in [13], it is possible to exploit a decentralized procedure that allows each agent to obtain an estimate of these values.

More specifically, as first introduced in [17], the estimation of λ_2 is computed by exploiting the estimation of the corresponding eigenvector v_2 . Specifically, the power iteration procedure described in [16] is used to design the following update law:

$$\begin{aligned} \dot{\tilde{v}}_2 = & -k_1 \text{Ave}(\{\tilde{v}_2^i\}) \mathbf{1} - k_2 L \tilde{v}_2 - k_3 \left(\text{Ave}(\{(\tilde{v}_2^i)^2\}) - 1 \right) \tilde{v}_2 \end{aligned} \quad (7)$$

where $k_1, k_2, k_3 > 0$ are the control gains, and $\text{Ave}(\cdot)$ is the averaging operation. Furthermore, $\tilde{v}_2^i$ is defined as the i -th agent's estimate of v_2^i , the i -th component of the eigenvector v_2 , and $\tilde{v}_2 = [\tilde{v}_2^1, \dots, \tilde{v}_2^N]^T$. Further details can be found in [17].

To implement the update law in Eq. (7) in a decentralized way, the averaging operation is implemented by means of the PI average consensus estimator described in [7].

Since there are two averaging operations in the update law in Eq. (7), two PI consensus estimators must be run:

- the first one, whose input is $\tilde{v}_2^i$, provides z_1^i as the i -th agent's estimate of $\text{Ave}(\{\tilde{v}_2^i\})$;
- the second one, whose input is $(\tilde{v}_2^i)^2$, provides z_2^i as the i -th agent's estimate of $\text{Ave}(\{(\tilde{v}_2^i)^2\})$.

Thus, each agent can run the decentralized version of the update law in Eq. (7):

$$\dot{\tilde{v}}_2^i = -k_1 z_1^i - k_2 \sum_{j \in \mathcal{N}_i} a_{ij} (\tilde{v}_2^i - \tilde{v}_2^j) - k_3 (z_2^i - 1) \tilde{v}_2^i \quad (8)$$

Let $\tilde{\lambda}_2$ be the value that the second smallest eigenvalue of the Laplacian matrix would take if $\tilde{v}_2$ were the corresponding eigenvector. As proved in [17], $\tilde{\lambda}_2$ can be computed as follows:

$$\tilde{\lambda}_2 = \frac{k_3}{k_2} [1 - \text{Ave}(\{(v_2^i)\})] \quad (9)$$

As shown in [17], $\frac{\partial \tilde{\lambda}_2}{\partial p_i}$ can be computed as follows

$$\frac{\partial \tilde{\lambda}_2}{\partial p_i} = \tilde{v}_2^T \frac{\partial L}{\partial p_i} \tilde{v}_2 = \sum_{j \in \mathcal{N}_i} \frac{\partial a_{ij}}{\partial p_i} (\tilde{v}_2^i - \tilde{v}_2^j)^2 \quad (10)$$

Then, from the definition of the edge-weights a_{ij} given in Eq. (5):

$$\frac{\partial \tilde{\lambda}_2}{\partial p_i} = \sum_{j \in \mathcal{N}_i} -a_{ij} (\tilde{v}_2^i - \tilde{v}_2^j)^2 \frac{p_i - p_j}{\sigma^2} \quad (11)$$

Further details can be found in [17].

The actual value of λ_2 can not be computed by each agent. In fact, the real value of $\text{Ave}(\{(v_2^i)\})$ is not available. Nevertheless, an estimate of this average, namely z_2^i , is available to each agent. Hence, the i -th agent can compute its own estimate of λ_2 , namely λ_2^i , as follows:

$$\lambda_2^i = \frac{k_3}{k_2} (1 - z_2^i) \quad (12)$$

As shown in [13], λ_2^i is a good estimate of both λ_2 and $\tilde{\lambda}_2$. More specifically, it has been proved that $\exists \Xi, \Xi' > 0$ such that

$$\begin{cases} |\lambda_2 - \lambda_2^i| \leq \Xi & \forall i = 1, \dots, N \\ |\tilde{\lambda}_2 - \lambda_2^i| \leq \Xi' & \forall i = 1, \dots, N \end{cases} \quad (13)$$

From Eq. (13), we can conclude that

$$|\lambda_2 - \tilde{\lambda}_2| \leq \Xi + \Xi' \quad (14)$$

The control law introduced in Eq. (4) will then be implemented introducing each agent's estimates, i.e.:

$$u_i^c = \text{csch}^2(\lambda_2^i - \tilde{\epsilon}) \frac{\partial \tilde{\lambda}_2}{\partial p_i} \quad (15)$$

where the value $\tilde{\epsilon}$ will be properly defined in the next section.

IV. CONNECTIVITY MAINTENANCE IN PRESENCE OF AN EXTERNAL CONTROLLER

In this section we extend the control law described in the previous section, considering the following control law:

$$\dot{p}_i = u_i^c + u_i^d \quad (16)$$

where u_i^c is the control term introduced in Eq. (15), while u_i^d is a control term used to obtain some desired behavior. Namely, the control term u_i^d is an unknown bounded function, i.e. $\|u_i^d\| \leq u_M$.

As in [13], we define the following energy function:

$$V(p) = \coth(\tilde{\lambda}_2 - \tilde{\epsilon}) \quad (17)$$

The energy function (Fig. 1(b)) is non-increasing (with respect to λ_2) and non-negative, for any $\tilde{\lambda}_2 > \tilde{\epsilon}$.

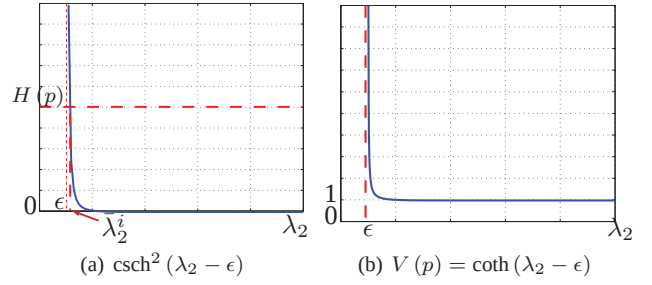


Fig. 1. Energy function and its derivative, with respect to λ_2

Proposition 1 Consider the dynamical system described by Eq. (16). Let Ξ, Ξ' be defined according to Eq. (13). $\exists \epsilon, \tilde{\epsilon}$ such that, if the initial value of $\lambda_2 > \tilde{\epsilon} + \Xi + \Xi'$, then the control law in Eq. (15) ensures that the value of λ_2 never goes below ϵ .

Proof: To prove the statement, we compute the time derivative of the energy function introduced in Eq. (17). From Eq. (17), it follows that:

$$\frac{\partial V}{\partial p_i} = \frac{\partial V}{\partial \tilde{\lambda}_2} \frac{\partial \tilde{\lambda}_2}{\partial p_i} = -\text{csch}^2(\tilde{\lambda}_2 - \tilde{\epsilon}) \frac{\partial \tilde{\lambda}_2}{\partial p_i} \quad (18)$$

From Eqs. (16), (15), (18), the time derivative of $V(p)$ can be computed as follows:

$$\begin{aligned} \dot{V}(p) &= \nabla_p V(p)^T \dot{p} = \sum_{i=1}^N \frac{\partial V}{\partial p_i} \dot{p}_i = \\ &= \sum_{i=1}^N \left[-\text{csch}^2(\tilde{\lambda}_2 - \tilde{\epsilon}) \frac{\partial \tilde{\lambda}_2}{\partial p_i} \right] \left[\text{csch}^2(\lambda_2^i - \tilde{\epsilon}) \frac{\partial \tilde{\lambda}_2}{\partial p_i} + u_i^d \right] \end{aligned} \quad (19)$$

Since u_i^d is bounded:

$$\begin{aligned} \dot{V}(p) &\leq \text{csch}^2(\tilde{\lambda}_2 - \tilde{\epsilon}) \sum_{i=1}^N \left[-\text{csch}^2(\lambda_2^i - \tilde{\epsilon}) \left\| \frac{\partial \tilde{\lambda}_2}{\partial p_i} \right\|^2 + \right. \\ &\quad \left. + \left\| \frac{\partial \tilde{\lambda}_2}{\partial p_i} \right\| u_M \right] \end{aligned} \quad (20)$$

From Eq. (20) it follows that $\dot{V}(p) \leq 0$ if

$$\sum_{i=1}^N \left[\text{csch}^2(\lambda_2^i - \tilde{\epsilon}) \left\| \frac{\partial \tilde{\lambda}_2}{\partial p_i} \right\|^2 \right] \geq u_M \sum_{i=1}^N \left\| \frac{\partial \tilde{\lambda}_2}{\partial p_i} \right\| \quad (21)$$

According to Eq. (13), the fact that the initial value of λ_2 is greater than $\tilde{\epsilon} + \Xi + \Xi'$ ensures that the initial value of λ_2^i is greater than $\tilde{\epsilon}$, $\forall i = 1, \dots, N$. Then, $\text{csch}^2(\lambda_2^i - \tilde{\epsilon})$ is monotonically decreasing. Hence,

$$\text{csch}^2(\lambda_2^i - \tilde{\epsilon}) \geq \text{csch}^2(\lambda_2^{MAX} - \tilde{\epsilon}) \quad (22)$$

where, according to Eq. (13)

$$\lambda_2^{MAX} = \max_{i=1, \dots, N} \{\lambda_2^i\} \leq \tilde{\lambda}_2 + \Xi' \quad (23)$$

Hence, according to Eqs. (22), (23) the condition in Eq. (21) is verified if

$$\text{csch}^2(\tilde{\lambda}_2 + \Xi' - \tilde{\epsilon}) \sum_{i=1}^N \left\| \frac{\partial \tilde{\lambda}_2}{\partial p_i} \right\|^2 \geq u_M \sum_{i=1}^N \left\| \frac{\partial \tilde{\lambda}_2}{\partial p_i} \right\| \quad (24)$$

If the following condition holds

$$\sum_{i=1}^N \left\| \frac{\partial \tilde{\lambda}_2}{\partial p_i} \right\|^2 \neq 0 \quad (25)$$

then the inequality in Eq. (24) can be rewritten as follows:

$$\text{csch}^2(\tilde{\lambda}_2 + \Xi' - \tilde{\epsilon}) \geq u_M \frac{\sum_{i=1}^N \left\| \frac{\partial \tilde{\lambda}_2}{\partial p_i} \right\|^2}{\sum_{i=1}^N \left\| \frac{\partial \tilde{\lambda}_2}{\partial p_i} \right\|} = H(p) > 0 \quad (26)$$

which implies

$$\tilde{\lambda}_2 \leq \bar{\lambda}_2 = \text{settcsh}\left(\sqrt{H(p)}\right) + \epsilon' \quad (27)$$

where $\epsilon' = \tilde{\epsilon} - \Xi'$. Therefore, as shown in Fig. 1(a), $\bar{\lambda}_2 > \epsilon'$ always exists that satisfies the condition in Eq. (27). Thus, $\forall \tilde{\lambda}_2 \leq \bar{\lambda}_2$ the function $\dot{V}(p) \leq 0$. Then, $\forall \tilde{\lambda}_2 \leq \bar{\lambda}_2$, the energy function $V(p)$ does not increase over time.

Let $\tilde{\lambda}_2(0) > \tilde{\epsilon}$ be the initial value of $\tilde{\lambda}_2$. If $\tilde{\lambda}_2(0) > \bar{\lambda}_2$, then the value of $\tilde{\lambda}_2$ will always be lower-bounded by $\bar{\lambda}_2$. Conversely, if $\tilde{\lambda}_2(0) \leq \bar{\lambda}_2$, then the value of $\tilde{\lambda}_2$ will increase, until $\tilde{\lambda}_2 \geq \bar{\lambda}_2$, and then it will never go below $\bar{\lambda}_2$. Namely, let

$$\hat{\lambda}_2 = \min(\tilde{\lambda}_2(0), \bar{\lambda}_2) \quad (28)$$

Then, the value of $\tilde{\lambda}_2$ will never go below the value of $\hat{\lambda}_2 > \epsilon'$.

In case $\left\| \frac{\partial \tilde{\lambda}_2}{\partial p_i} \right\| = 0$, the condition in Eq. (25) is not verified. However, in this case

$$\dot{\tilde{\lambda}}_2 = \frac{\partial \tilde{\lambda}_2}{\partial p_i} \dot{p}_i = 0 \quad (29)$$

This implies that the value of $\tilde{\lambda}_2$ is constant over time, thus it is lower-bounded by its initial value. In both cases, $\tilde{\lambda}_2 > \epsilon'$.

Then, according to Eq. (14), the control law in Eq. (15) ensures that the value of λ_2 never goes below $\epsilon = \epsilon' - \Xi = \tilde{\epsilon} - \Xi' - \Xi$. ■

Hence, the fact that the estimation errors are bounded allows us to prove that, by ensuring an appropriate lowerbound on the estimate of λ_2 , we guarantee the actual value of λ_2 to be strictly greater than zero.

V. RENDEZVOUS AND FORMATION CONTROL

In this section we exploit the control strategy introduced in Section III and Section IV to address the connectivity maintenance issue within rendezvous and formation control

problems. For this purpose, we first rewrite the connectivity maintenance control action in vector form. Let

$$\bar{a}_{ij} = \text{csch}^2(\lambda_2^i - \tilde{\epsilon}) \frac{1}{\sigma^2} (\tilde{v}_2^i - \tilde{v}_2^j)^2 a_{ij} \quad (30)$$

Furthermore, let the modified degree matrix of the graph be defined as $\bar{D} = \text{diag}(\{\bar{d}_i\})$, where $\bar{d}_i$ is the degree of the i -th node of the graph, i.e. $\bar{d}_i = \sum_{j=1}^N \bar{a}_{ij}$. The modified Laplacian matrix can be then defined accordingly, as $\bar{L} = \bar{D} - \bar{A}$.

We now introduce the component-wise operator, as defined in [9]:

$$c(\xi, k) = [\xi_{1,k} \dots \xi_{N,k}]^T \quad k = 1, \dots, m \quad (31)$$

where $\xi \in \mathbb{R}^m$. For instance, since $p_i \in \mathbb{R}^m$, the vector $c(p, k)$ collects the k -th components of the positions of the agents.

Hence, the control law in Eq. (16) can be rewritten in vector form, component-wise:

$$\frac{d}{dt} c(p, k) = -\bar{L} c(p, k) + c(u^d, k) \quad k = 1, \dots, m \quad (32)$$

where u^d is the vector containing the desired control laws.

A. Consensus-based rendezvous

In this section we address the connectivity maintenance issue in the rendezvous problem, i.e. making a group of agents converge to the same position. We then introduce the following consensus-based control law:

$$c(u^d, k) = -L_* c(p, k) \quad k = 1, \dots, m \quad (33)$$

As shown in [12], this control law guarantees the convergence of the system to the rendezvous configuration, provided that the communication graph is connected.

Proposition 2 *Under the control law described by Eqs. (32), (33), the connectivity of the communication graph is always ensured.*

Proof: To prove the statement, we will exploit the results given in Proposition 1. More specifically, we will now find the value u_M to be used in Eqs. (26), (27) to define the lowerbound on λ_2 . Since

$$\begin{aligned} c(u_i^d, k) &= c\left(\sum_{j \in \mathcal{N}_i} (p_i - p_j), k\right) \quad k = 1, \dots, m \\ &= \sum_{j \in \mathcal{N}_i} c((p_i - p_j), k) \quad k = 1, \dots, m \end{aligned} \quad (34)$$

we can conclude that

$$\|u_i^d\| \leq u_M = m \max_{k=1, \dots, m} |c(u_i^d, k)| \leq mNR \quad (35)$$

since R is the greatest possible distance between two neighboring agents. ■

Proposition 3 *Under the control strategy introduced in Eqs. (32), (33), the system asymptotically converges to the rendezvous configuration.*

Proof: The proof is analogous to that of [12], considering $\tilde{L} = \bar{L} + L_*$ as the Laplacian matrix of the communication graph. ■

B. Consensus-based formation control

In this section we will address the connectivity maintenance issue in the formation control problem. Consensus-based control laws can be exploited for formation control as well. In [5] the following control strategy has been introduced:

$$c(u^d, k) = -L_* c(p, k) + c(b, k) \quad k = 1, \dots, m \quad (36)$$

where

$$c(b, k) = L_* c(\bar{p}, k) \quad k = 1, \dots, m \quad (37)$$

and $\bar{p}$ is a vector containing the desired positions for the agents in the formation. More specifically, $\bar{p}$ represents the desired relative position of each agent with respect to the center of the formation. Hence, the desired configuration can be described as follows:

$$c(p, k) = \alpha_k \mathbf{1} + c(\bar{p}, k) \quad k = 1, \dots, m \quad (38)$$

for some $\alpha_k \in \mathbb{R}$, $\forall k = 1, \dots, m$. In other words,

$$c(p, k) - c(\bar{p}, k) = c(p - \bar{p}, k) \in \text{span}(\mathbf{1}) \quad k = 1, \dots, m \quad (39)$$

As in the rendezvous case, the convergence of the system to the desired configuration is guaranteed, provided that the communication graph is connected.

Proposition 4 *Under the control law described by Eqs. (32), (36), the connectivity of the communication graph is always ensured.*

Proof: The proof is analogous to that of Proposition 2. ■

In [5] the proof of the convergence of the system to the desired formation is provided. We will now show how to extend this proof, in order to guaranteed the convergence even in presence of the connectivity maintenance control action.

Let $\tilde{L} = \bar{L} + L_*$. From Eqs. (32) and (36) we obtain

$$\frac{d}{dt} c(p, k) = -\tilde{L} c(p, k) + L c(\bar{p}, k) \quad k = 1, \dots, m \quad (40)$$

It's easy to show that $\tilde{L}$ is a well defined weighted Laplacian matrix for the communication graph.

Let $\Gamma \in \mathbb{R}^{N \times N}$ be a matrix such that $L = \tilde{L}\Gamma$, and let $q \in \mathbb{R}^N$ be defined such that

$$c(q, k) = c(\Gamma \bar{p}, k) \quad k = 1, \dots, m \quad (41)$$

Thus, Eq. (40) can be rewritten as follows:

$$\begin{aligned} \frac{d}{dt} c(p, k) &= -\tilde{L} c(p, k) + \tilde{L} \Gamma c(\bar{p}, k) \\ &= -\tilde{L} c(p, k) + \tilde{L} c(q, k) \\ &= -\tilde{L} c(p - q, k) \quad k = 1, \dots, m \end{aligned} \quad (42)$$

The control action described in Eq. (42) is zero if and only if one of the following conditions is verified:

- 1) $p = q$,
- 2) $c(p - q, k) \in \ker(L)$, $\forall k = 1, \dots, m$.

Since $\ker(L) = \text{span}(\mathbf{1})$, these conditions are verified when

$$c(p - q, k) = \alpha_k \mathbf{1} \quad k = 1, \dots, m \quad (43)$$

for some $\alpha_k \in \mathbb{R}$. Hence,

$$c(p, k) = c(q, k) + \alpha_k \mathbf{1} \quad k = 1, \dots, m \quad (44)$$

It's reasonable to assume that, in the desired configuration, the value of λ_2 is (sufficiently) greater than ϵ . Thus, we can assume that, in a neighborhood of the desired configuration, the magnitude of the connectivity maintenance control action is negligible. This implies that $\bar{L} \approx 0_N$, where 0_N is a n -by- n matrix whose elements are all equal to zero. Thus, in this case $\tilde{L} \approx L$, that implies $\Gamma \approx I_N$, where I_N is the identity matrix of size N . This implies that $q \approx \bar{p}$: the convergence to the desired configuration is then guaranteed.

VI. SIMULATIONS AND EXPERIMENTS

A. Simulations

In order to validate the control strategy presented in this paper, we implemented several Matlab simulations. We have varied the number of agents, from $N = 3$ to $N = 20$, and we have placed them in randomly chosen initial positions on a bidimensional environment.

As shown in the accompanying video clip, we performed the simulations both for the rendezvous and for the formation control case.

The first part of the video clip shows a rendezvous simulation, with six point mass agents that, starting from random initial positions, are supposed to converge to a common point, while avoiding collisions with randomly placed point obstacles (blue dots in the video). For obstacle avoidance purposes, a repulsive artificial potential field has been added (see e.g. [15]), ensuring that it produces a bounded control action. Without the connectivity maintenance controller, the connectivity is lost quite soon: the obstacle avoidance action obstructs the desired movement of some agents, that are thus trapped and lose connectivity with the other ones. As expected, using the connectivity maintenance control action, since all the additional control laws are bounded, the connectivity of the graph is always preserved, as shows also in Fig. 2(a).

The second part of the video clip shows a formation control simulation, with six point mass agents that, starting from random initial positions, are supposed to converge to an hexagonal formation, and move at constant velocity along the x -axis, while avoiding collisions with randomly placed point obstacles. To make the formation move in a desired direction, we added a common offset to the control law in Eq. (36), that describes the desired speed of the barycenter of the formation.

As in the rendezvous case, without the connectivity maintenance controller, the connectivity is lost quite soon: the

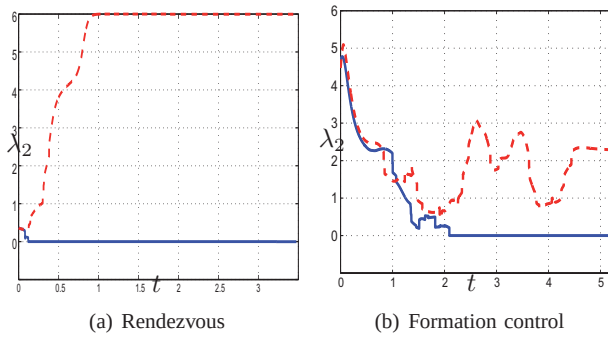


Fig. 2. Simulation results: value of λ_2 with (red dashed line) and without (blue solid line) the connectivity maintenance controller

obstacle avoidance action obstructs the desired movement of some agents, that are thus trapped and lose connectivity with the other ones. As expected, using the connectivity maintenance control action the connectivity of the graph is always preserved, as shows also in Fig. 2(b).

B. Experiments

Experiments are performed within the software platform described in [1]. A group of four E-puck robots [11] are moving in a $2.0m \times 1.5m$ arena. The position of each robot is monitored with a webcam positioned over the arena, and the robots are tracked using colored markers. The control strategy is implemented on a central PC that controls the robots exploiting bluetooth communication. The control software is written to emulate a decentralized implementation, and the presence of limited communication capabilities.

E-puck robots can be described by the differential-drive kinematic model. To deal with the fact that this model represents a nonholonomic system, we applied the feedback linearization technique presented in [14].

The last part of the accompanying video clip shows the system implementing a rendezvous task. The robots, starting from random initial positions, are supposed to converge to a common point, while avoiding collisions with three point obstacles (represented with black crosses in the video clip). Without the connectivity maintenance controller, the connectivity is lost quite soon: the obstacle avoidance action drives one of the robots too far away from the other ones, thus causing loss of connectivity. As expected, using the connectivity maintenance control action the connectivity of the graph is always preserved, and the robots converge to the rendezvous configuration.

VII. CONCLUSIONS

In this paper we have introduced a control strategy that ensures connectivity maintenance in presence of an external bounded control term. Specifically, we have exploited this control strategy to address the connectivity maintenance issue within rendezvous and formation control applications.

Matlab simulations and experiments on real robots have been implemented for validation purposes. Due to the limited size of the experimental arena, we have not been able

to extensively test our algorithm in the formation control application. In the future, we will implement the control algorithm on robots moving in a wider arena, in order to be able to reproduce the scenario represented in the simulations: several obstacles, randomly placed in the environment, that can cause disconnection of the communication graph as the robots move.

Future work aims at implementing the connectivity maintenance algorithm for groups of dynamic agents, instead of considering only simple integrators. Furthermore, we will try to extend our algorithm to more complex communication models, possibly including the presence of time delays.

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